

Evaluating Multimodal LLM Extraction for Vector Trait Curation: High Precision with Structure-Dependent Record Recovery

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Declaration

I declare that this dissertation reports my own research and analysis, except where contributions from others are stated below. The source material consisted of an existing VecTraits export and published scientific papers. I selected 20 papers from the VecTraits export and manually curated the development and held-out reference benchmarks used in this dissertation, comprising a total of 2,766 data rows. I designed, implemented, and ran the extraction and evaluation workflow, and wrote all code used for data processing, extraction, evaluation, quantitative analysis, and generation of the reported results. I also carried out the data cleaning and qualitative discrepancy review, and wrote the dissertation.

Dr Lauren Cator and Professor Samraat Pawar provided supervisory guidance on project design, interpretation, and presentation of the work. Stanislav Modrak provided advice and assistance with the computational implementation. The development benchmark was additionally reviewed by Sarah Kelly.

The large language model evaluated in this project, Gemma 4 31B IT, was an existing open-source model accessed through a hosted API and was not developed by me or by my supervisors. The extraction schema, prompting workflow and evaluation framework were implemented for this project under supervisory guidance. Any use of generative AI in preparing the dissertation is disclosed separately on the GenAI Use Disclosure page.

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GenAI Use Disclosure

Generative AI tools may support aspects of learning; transparency about their use is required by the Department of Life Sciences. The disclosures below record how generative AI was used in preparing this project report.

Option	Form description	Disclosure for this project
☐ No AI Use	No AI tools were used.	Not applicable.
☒ Language and Editing	Used for spelling, grammar, clarity, or formatting.	ChatGPT was used to suggest improvements to grammar, clarity and LaTeX formatting.
☒ Concepts and Planning	Used to explain concepts, brainstorm ideas, or help outline/organise the work.	ChatGPT was used to discuss concepts and help plan the structure of sections.
☒ Technical and Analytical Assistance	Used for programming support, debugging, suggesting analytical approaches, interpreting outputs or explaining statistical results.	ChatGPT and Claude were used for debugging support.
☐ Content and Media Generation	Used to generate draft text, figures, tables or legends as ideas for content.	None declared
☐ Other	Other use not described above.	None declared.

I confirm that any use of genAI tools is in accordance with Departmental guidelines on genAI use and it has been disclosed. All submitted work has been reviewed and edited by me, reflects my own understanding, and no content generated by AI has been presented as my own work. I take full responsibility for the academic integrity of the final submission.

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Abstract

Vector-borne disease research depends on trait data reported across a large and heterogeneous literature, but manually extracting them from PDFs is slow and difficult to scale. This project developed and evaluated a reproducible, provenance-aware pipeline using a multimodal large language model (Gemma 4 31B IT) to extract original vector trait records from scientific PDFs for VecTraits curation. A diversity-oriented 20-paper corpus comprised 15 development papers (2,335 reference records) and five held-out papers (431 records). Records were extracted under a reduced, MIREVTD-aligned 22-field schema and evaluated using bipartite record matching followed by row-set, conditional field and end-to-end field metrics. Across 11 prompt versions, v10 achieved the highest development conditional macro field F1 (0.810) and was selected before held-out evaluation because the intended curation workflow prioritised stronger field agreement within a smaller candidate set over broader extraction requiring more correction. On the held-out papers, v10 achieved row precision of 1.000, row recall of 0.360, conditional macro field F1 of 0.729 and end-to-end macro field F1 of 0.394. The held-out results showed high row precision but incomplete, structure-dependent record recovery, while conditional field agreement was lower than on the development papers. The pipeline is therefore best suited to generating candidate records for human review, with field verification and completeness checking retained as explicit curation steps.

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1 Introduction

Vector-borne diseases (VBDs) remain a persistent and expanding challenge for human, animal and plant health, and their dynamics are strongly shaped by the biological traits of vectors. Here, a trait is defined as any measurable morphological, physiological, behavioural or life-history characteristic of an organism that can influence its fitness or its interactions with the environment (Violle et al., 2007). Traits of vectors such as development time, survival, longevity, fecundity, feeding behaviour and transmission-related characteristics affect not only pathogen transmission, but also vector abundance, life history and responses to environmental change (Cator et al., 2020). Trait-based approaches are therefore increasingly central to understanding variation in disease transmission and to building ecological and epidemiological models of vector-borne disease systems.

Despite their importance, vector trait data remain difficult to synthesise and reuse at scale. One major obstacle is heterogeneous reporting across the literature: trait names differ between studies, definitions are often incomplete or inconsistent, units may be omitted or expressed differently, and information may be distributed across text, tables and figures. Such inconsistencies hinder data discoverability, interpretation and integration, reflecting broader semantic challenges recognised across ecological data systems (Madin et al., 2008). As a result, vector trait data often do not fully satisfy the FAIR principles of being findable, accessible, interoperable and reusable (Wilkinson et al., 2016).

VecTraits is a repository of trait data for vectors, pathogens and hosts relevant to VBDs (Johnson et al., 2023). The VecTraits export used in this study, downloaded on 27 April 2026, contained 334 paper-level units and over 40,000 rows of trait data. In this export, some fields were not consistently populated: original trait name and original trait definition did not always preserve source-paper wording, while units, life stage, experimental conditions and error terms could be incomplete or inconsistently recorded. MIREVTD, a Minimum Information standard for Reporting Vector Trait Data, addresses such inconsistencies by defining minimum reporting information intended to improve the consistency and reusability of vector trait data (Ryan et al., 2026). However, MIREVTD primarily guides future reporting and does not address the recovery of information from existing papers, where trait names, definitions, units and contex-

tual information remain heterogeneous.

Therefore, a practical intermediate workflow is needed for existing literature. Trait data can first be recovered as originally reported, together with information linking each value to its source, before later harmonisation to broader standardised trait names or definitions. In this study, a record refers to one paper-reported trait measurement together with its trait description, value, unit, organism, experimental condition and source information. The workflow therefore focused on recovering source-level records while preserving their provenance. This provenance is important for human-in-the-loop curation because each extracted value remains traceable to the text, table or figure from which it was recovered for expert verification.

Automated information extraction offers one possible way to operationalise this source-recovery step. In broader scientific information extraction, both task-specific fine-tuning and prompt-based LLM approaches have achieved strong performance for extracting structured data from scientific text (Dagdelen et al., 2024; Polak and Morgan, 2024), while schema-driven LLM approaches have also performed well on heterogeneous scientific tables (Bai et al., 2024). This progress is also extending into ecology. Prompt-based LLMs have been used to extract ecological information such as species occurrences and locations from unstructured text, although performance varies across information types and can decline for more complex quantitative variables (Gougherty and Clipp, 2024; Castro et al., 2024). Trait-specific studies provide further evidence that automated extraction is feasible. FloraTraiter used a largely rule-based NLP workflow to recover computable plant traits from descriptive biodiversity literature (Folk et al., 2024), while a more recent transformer-based pipeline achieved mean precision of 90.8% and recall of 88.6% across five categorical plant traits and also evaluated numerical-trait extraction from unstructured species descriptions (Domazetoski et al., 2025). Together, these studies show that substantial scientific, ecological and trait information can be recovered automatically, particularly from text-centred or structurally constrained sources.

Despite this progress, important gaps remain for vector-trait curation. Existing vector-trait resources still rely heavily on expert-led literature extraction. For example, AedesTraits compiled published mosquito trait measurements manually and used figure digitisation when

raw data were unavailable (Da Re et al., 2025). Automated LLM-based extraction of complete vector-trait records remains largely unevaluated, while the task extends beyond many existing text-centred extraction problems. A single vector-trait record may require information distributed across prose, tables and figures, together with organismal context, experimental conditions and source provenance. Full-document multimodal extraction also remains challenging. MatViX evaluated vision-language models on 324 full-length scientific articles with structured outputs spanning text, tables and figures and found substantial room for improvement in zero-shot extraction (Khalighinejad et al., 2025). Human-in-the-loop systems such as SciDaSynth similarly combine multimodal extraction with explicit validation and refinement rather than treating model outputs as final data (Wang et al., 2025). The unresolved problem is therefore not simply whether LLMs can extract scientific information, but whether an off-the-shelf multimodal model can reconstruct complete, provenance-linked vector-trait records from heterogeneous PDFs accurately and consistently enough to support expert curation.

The overall objective of this project was to build and evaluate a reproducible, provenance-aware pipeline for extracting original vector trait metadata from scientific PDFs to support human-in-the-loop VecTraits curation. Five research questions guided the study: (1) to what extent can an off-the-shelf multimodal large language model recover schema-constrained vector-trait records from scientific PDFs, (2) how does iterative prompt engineering relate to changes in record recovery and field-level agreement, (3) which fields and document structures are associated with the main extraction errors and omissions, (4) how does the final pipeline perform on papers held out from prompt development compared with the development papers, and (5) within a human-in-the-loop curation workflow, what division of work between model extraction and curator review is supported by the observed performance? The project contributes a reproducible, provenance-aware extraction and evaluation pipeline, a systematic assessment of record recovery and field-level extraction performance, and an evidence-based framework for incorporating LLM-assisted extraction into ecological data curation. More broadly, source-faithful recovery of legacy trait data could support subsequent harmonisation and cross-study synthesis, strengthening the empirical basis for ecological and epidemiological modelling of vector-borne disease.

2 Methods

For this project, I developed a reproducible, human-in-the-loop workflow for recovering vector-trait metadata from scientific PDFs to support VecTraits curation. The workflow comprised corpus assembly, definition of a reduced extraction schema, reference-benchmark construction, multimodal input preparation and record extraction, output validation, and quantitative evaluation with qualitative discrepancy analysis (Figure 1).

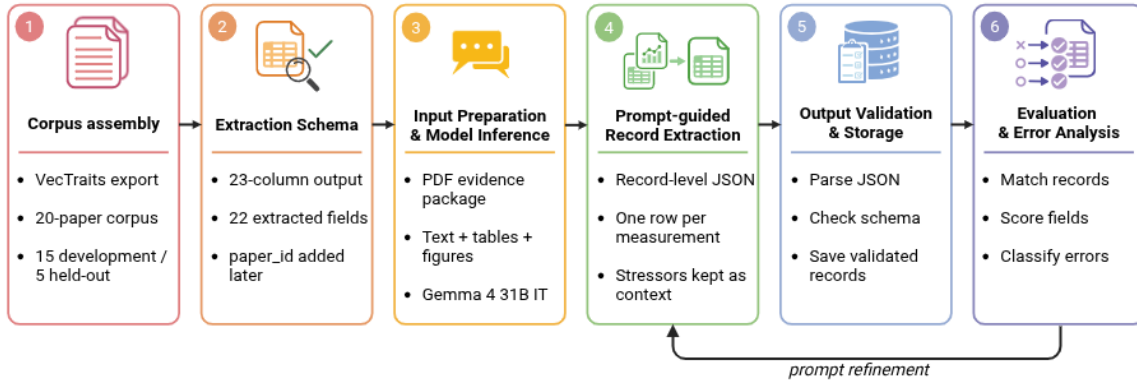


Figure 1: Overall workflow for multimodal LLM extraction and evaluation of vector-trait records from scientific PDFs. The backwards arrow indicates the development-only prompt-refinement loop, with the held-out benchmark outside this loop. Created with BioRender.com.

2.1 Corpus assembly

The source corpus was assembled from the VecTraits export downloaded on 27 April 2026. Records were grouped into paper-level units using DOI where available and citation otherwise, and assigned stable identifiers (e.g. `paper_001`) to link PDFs, model outputs, reference records and evaluation files. Papers whose relevant curated data were primarily contained in supplementary files or whose main PDF was unavailable were excluded because the study focused on values embedded in accessible main PDFs that require curator extraction.

Before selection, candidate papers were characterised by primary trait family, reporting format (text-heavy, table-heavy, figure-heavy or mixed), dataset source, citation group and existing VecTraits row count. A 20-paper corpus was selected because manual reference curation and discrepancy review were time-intensive. Although limited in size, the corpus was

designed to maximise diversity in biological trait types, reporting formats, dataset sources and citation groups, allowing the pipeline to be evaluated across heterogeneous extraction conditions rather than as a probability sample of the wider VecTraits literature. Selection began with the candidate containing the largest number of existing VecTraits rows within each primary trait family, followed by the highest-row remaining candidate for each reporting format not yet represented. The remaining slots were filled greedily. At each step, the candidate contributing the greatest additional weighted coverage was selected, the represented corpus features were updated, and scores were recalculated for the remaining papers (Nemhauser et al., 1978; Cormen et al., 2009).

$$\begin{aligned} \text{Score}_i = & 4 \times \text{NewPrimaryFamily}_i + 2 \times \text{AdditionalFamilies}_i + 3 \times \text{NewFormat}_i \\ & + 1.5 \times \text{NewDataset}_i + 1.5 \times \text{NewCitation}_i + 0.75 \times \text{Size}_i. \end{aligned} \quad (1)$$

Here, $\text{NewPrimaryFamily}_i$, NewFormat_i , NewDataset_i and NewCitation_i indicated whether paper i added a category not already represented in the selected corpus. $\text{AdditionalFamilies}_i$ counted further new trait-family labels, while Size_i was the number of existing VecTraits rows associated with paper i , capped at 40 and divided by 40. The weights were design choices rather than estimated statistical parameters. Trait family and reporting format received the greatest weights because biological and document-structure diversity were the main sampling priorities, whereas dataset source, citation group and row count were used to reduce redundancy and retain informative larger papers without allowing size to dominate selection.

After the 20-paper corpus was selected, it was divided into 15 development papers and five held-out papers, with the held-out subset selected using the same greedy diversity procedure. Only the main PDF of each included paper was used for extraction. The development papers were used for iterative prompt engineering, extraction-rule refinement and review workflow testing, and comprised nine table-heavy, three mixed-format and three figure-heavy papers. Per-paper primary trait family and reporting format are provided in Appendix Table A1.

2.2 Extraction schema

The full 91-column VecTraits schema was not used because many fields were database-specific, standardised or highly optional, and requiring such a large output space would add complexity to record-level extraction. A reduced schema guided by MIREVTD was therefore defined for both reference curation and model output (Ryan et al., 2026). It contained 23 columns: `paper_id`, added by the pipeline to link each row to its source paper, and 22 fields targeted for model extraction. Each row represented one paper-reported measurement, with its trait description, value, unit, uncertainty, organism, experimental condition and PDF source kept together as a single record. Fields were left empty when the corresponding information was not clearly reported.

Experimental and environmental conditions were represented using the shared `stressor`, `stressor_def`, `stressor_value` and `stressor_unit` fields rather than separate condition-specific fields. Where multiple conditions applied to a record, the main condition was captured in these fields and additional context or ambiguity was recorded in `notes`. The field groups and their inclusion rationale are summarised in Table 1, with full field definitions and examples provided in Appendix Table B1.

Table 1: Reduced MIREVTD-aligned schema used for reference curation and model output.

MIREVTD-related information	Extraction field(s)	Reason for inclusion
Study context	<code>location</code> , <code>lab_field</code>	Records where the study was carried out and whether the measurement came from laboratory or field conditions.
Trait description	<code>original_trait_name</code> , <code>original_trait_def</code>	Identifies the measured trait and records how the source paper defined or described it.
Trait value and unit	<code>original_trait_value</code> , <code>original_trait_unit</code>	Records the reported value and original unit needed to interpret and reuse the trait record.
Error, variation and sample size	<code>original_error_pos</code> , <code>original_error_neg</code> , <code>original_error_unit</code> , <code>interactor1number</code>	Captures reported uncertainty and the number of primary organisms used in the measurement, where available.

Continued on next page

Table 1: Reduced MIREVTD-aligned extraction schema continued.

MIREVTD-related information	Extraction field(s)	Reason for inclusion
Primary organism	<code>interactor1genus</code> , <code>interactor1species</code> , <code>interactor1stage</code> , <code>interactor1sex</code>	Identifies the organism being measured and records key biological details such as life stage and sex.
Second organism, where relevant	<code>interactor2genus</code> , <code>interactor2species</code>	Records another organism involved in the measured trait, such as a host, prey or pathogen.
Axes of variation	<code>stressor</code> , <code>stressor_def</code> , <code>stressor_value</code> , <code>stressor_unit</code>	Records the experimental or environmental condition under which the trait was measured, such as temperature, humidity, diet, insecticide exposure, host type or another treatment.
PDF provenance and extraction notes	<code>figure_table</code> , <code>notes</code>	Records where the value was reported in the PDF and any additional information needed to interpret the row.

2.3 Reference benchmark construction

For the 15 development papers, the original VecTraits export was not used as the quantitative reference because it did not provide a complete and consistently schema-aligned representation of the source papers for this evaluation. Some supported paper-reported records were absent, some unsupported rows were present, and original trait names, definitions and other fields were not always recorded consistently under the project-specific extraction rules. A manually curated development benchmark of 2,335 records was therefore constructed by re-curating the source papers under the reduced extraction schema (Table 1). The development benchmark was curated by the author and reviewed by a data-curation expert (S. Kelly, One Health VBD Hub Curator), with ambiguous cases resolved through discussion. The same schema was then used by the author to construct a separate held-out benchmark of 431 records from the five held-out main PDFs.

2.4 Input preparation and model inference

The principal extraction model was Gemma 4 31B IT (`google/gemma-4-31B-it`), an open, instruction-tuned multimodal model capable of processing text and image inputs and generating text outputs (Google AI for Developers, 2026). The 31B dense variant was selected because its 256K-token context window and multimodal capabilities supported paper-level inputs combining article text, tables, captions and figure evidence. During inference, the model generated schema-constrained JSON records through conditional next-token prediction guided by the extraction prompt, document evidence and target schema.

Because the model did not accept PDF files, each PDF was converted into textual and visual evidence. Machine-readable article text was extracted page by page with PyMuPDF, with OCR used only when a PDF lacked a usable text layer. Tables were processed with Docling using table-structure recognition to preserve row and column relationships. All detected table and figure regions were rendered at 150 DPI and supplied as visual evidence together with their captions and surrounding article text. Each model request contained at most 45,000 evidence characters and 12 images, and evidence packages exceeding either limit were split across multiple requests. The reported v1–v11 runs used the Gemini API alias `gemma-4-31b-it` between 23 June and 15 August 2026. Although the configuration file specified `temperature = 0` and `max_new_tokens = 8192`, these values were not passed to `generate_content` and were therefore not explicitly controlled in the reported runs. The implementation used PyMuPDF 1.27.2.3, Docling 2.107.0, pandas 3.0.3, SciPy 1.18.0 and `google-genai` 2.6.0.

2.5 Prompt-guided record extraction

Prompt versions were retained separately, and successive versions were revised after reviewing development-set extraction errors. Later prompts introduced more explicit record-level rules where earlier versions produced broad or fragmented outputs. The model was instructed to extract complete paper-reported measurement records rather than isolated trait-like phrases, with separate records required for measurements reported across different temperatures, species, stages, treatments or locations.

The model was instructed to use textual, tabular and visual evidence together. For figure-

derived data, it was asked to consider figure legends, axis labels, tick marks, units and article context, and to digitise plotted values only when they were clearly readable. Uncertain values were returned as missing and recorded in notes. Additional constraints required the model to preserve author-reported wording where possible, return unreported optional fields as null, avoid inventing values or units, and exclude background statements, malformed fragments and information belonging to other schema fields. The `figure_table` field recorded the primary table or figure source for each value, while climate-background tables and statistical-summary figures were excluded as extraction sources.

2.6 Output validation and storage

All prompts required JSON-only output following the target schema. Model responses were stripped of Markdown code fences where present, parsed as JSON and checked for a valid top-level `records` list and the expected field structure. Invalid JSON, a missing `records` list, failed schema validation or another processing exception triggered a paper-level retry, with up to five attempts allowed per paper and transient API errors retried separately up to four times. Where multiple input chunks were required, their extracted records were merged to form the paper-level output. Missing optional fields were filled with empty values. `original_trait_name` was treated as the minimum required identifier of a trait record, so records lacking this field were removed before evaluation. `paper_id` was then added from the corpus manifest or source filename, and the final validated records were stored separately for each run for downstream evaluation.

2.7 Evaluation and error analysis

Evaluation comprised one-to-one record alignment, row-set and field-level metrics, and qualitative discrepancy analysis (Figure 2).

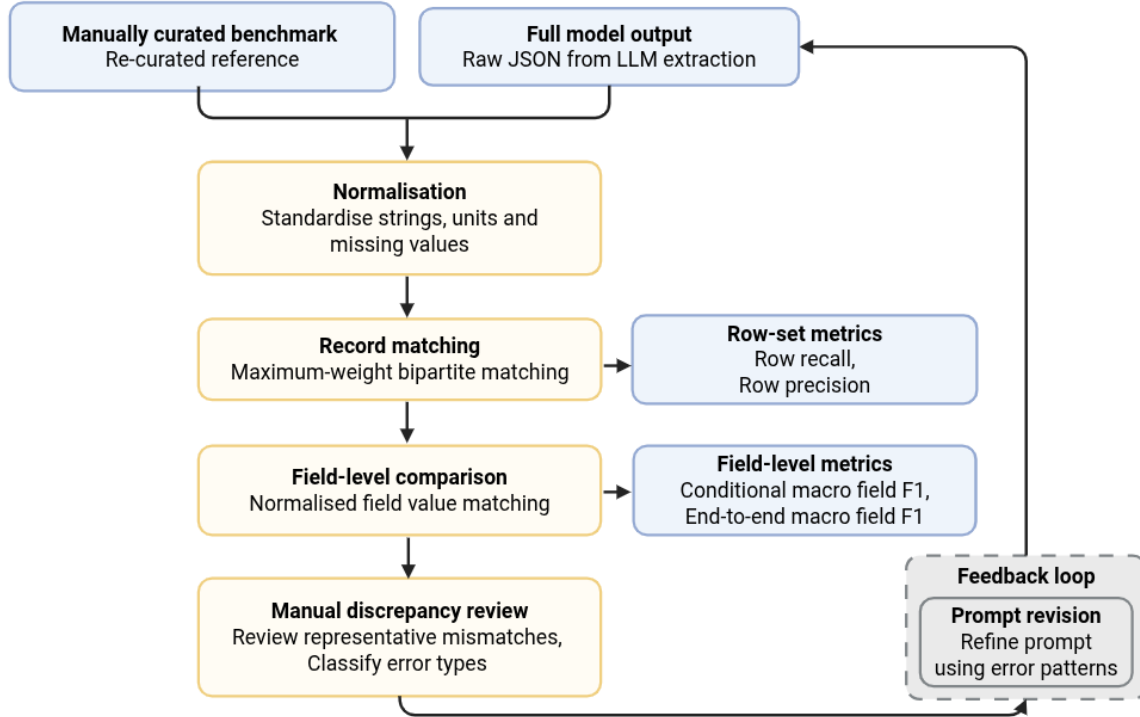


Figure 2: Evaluation workflow showing record normalisation, matching, quantitative evaluation and development-only discrepancy feedback. Blue boxes indicate inputs and quantitative outputs, and yellow boxes indicate evaluation steps.

2.7.1 Pre-matching normalisation

Before record alignment, model outputs and reference records were normalised field by field. This normalisation was applied for evaluation only and did not modify the original model-output or reference files. Normalisation converted text to lower case, standardised whitespace, mapped common missing-value strings to empty values, harmonised units through aliases, parsed numeric fields into comparable numeric values, and applied simple aliases to life stage, sex and study setting. The complete field-specific normalisation rules are provided in Appendix Table B2.

2.7.2 Record matching and row-set metrics

After field-specific normalisation, model records were aligned with reference records within each paper using one-to-one bipartite matching (Crouse, 2016). This was necessary because model and reference records could appear in different orders and could differ in number. Matching therefore identified the most compatible model–reference record pairs while allowing unmatched records to remain.

Eight schema fields were used to calculate a project-specific matching score. For these fields, punctuation was removed from the already normalised strings so that superficial formatting differences did not prevent otherwise identical values from matching. For each possible model–reference pair, let $I_j(m, r) = 1$ when field j was non-empty in both records and the resulting values were identical, and $I_j(m, r) = 0$ otherwise. The pair score was

$$S(m, r) = 3I_{\text{original_trait_name}}(m, r) + \sum_{j \in J} I_j(m, r), \quad (2)$$

where J comprised `original_trait_value`, `original_trait_unit`, `figure_table`, `interactor1species`, `interactor1stage`, `stressor_value` and `stressor_unit`. Scores ranged from 0 to 10. `original_trait_name` received weight 3 because it was intended to distinguish records more strongly than contextual fields that could be shared by multiple records within the same paper, while each remaining field received weight 1.

The resulting score matrix was passed to SciPy’s `linear_sum_assignment` function to obtain the maximum-weight one-to-one assignment. Assigned pairs were retained only when at least one matching field agreed. This was equivalent to the implementation threshold of 0.3 because pair scores were integer-valued, so zero-score pairs were rejected and all positive-score pairs were retained. No domain-specific tie-breaking rule was applied. Unmatched model records were classified as extra records, while unmatched reference records were classified as missing records.

Row-set precision and recall were calculated from the numbers of matched, extra and missing records:

$$\text{Row-set Precision} = \frac{n_{\text{matched}}}{n_{\text{matched}} + n_{\text{extra}}} \quad (3)$$

$$\text{Row-set Recall} = \frac{n_{\text{matched}}}{n_{\text{matched}} + n_{\text{missing}}} \quad (4)$$

Here, n_{matched} is the number of model records matched to reference records, n_{extra} is the number of unmatched model records, and n_{missing} is the number of unmatched reference records. Row-set precision therefore penalised over-extraction, while row-set recall penalised under-extraction. Row-set precision measures whether an extracted record could be aligned to a reference record under the predefined matching rule rather than whether every field within that record was correct. Field correctness was evaluated separately using the field-level metrics described below.

2.7.3 Field-value comparison and field metrics

After record matching, the value in each schema field was compared within every accepted model–reference record pair. Of the 23 output fields, `paper_id` was excluded because it was added by the pipeline rather than extracted by the model, and `notes` was excluded because it was used only for qualitative interpretation. Quantitative field-level evaluation therefore covered 21 fields.

For field scoring, most fields required equality after normalisation. Numeric trait values, error values and stressor values were accepted within 1% relative tolerance, whereas `interactor1number` required exact numeric agreement. For `location`, `original_trait_def` and `stressor_def`, a non-exact pair was accepted when one normalised value contained the other or when the values shared at least two words of four or more characters. This relaxed rule accommodated shorter and longer descriptions but could also accept superficially related phrases, so results for these fields are interpreted as potentially optimistic.

For each scored field j , a true positive TP_j was recorded when both values were non-empty and matched under the relevant field-comparison rule. A model value paired with an empty reference value contributed a false positive FP_j , and a reference value paired with

an empty model value contributed a false negative FN_j . A non-matching pair of non-empty values contributed both one FP_j and one FN_j . Cases where both values were empty were recorded as true negatives but excluded from precision, recall and F1.

Field precision, recall and F1 were calculated separately for each scored field, following established use of these metrics in schema-constrained information extraction (Swain and Cole, 2016; Yao et al., 2019):

$$\text{Precision}_j = \frac{TP_j}{TP_j + FP_j} \quad (5)$$

$$\text{Recall}_j = \frac{TP_j}{TP_j + FN_j} \quad (6)$$

$$F1_j = \frac{2 \times \text{Precision}_j \times \text{Recall}_j}{\text{Precision}_j + \text{Recall}_j} \quad (7)$$

For conditional field metrics, TP_j , FP_j and FN_j were counted only across accepted matched record pairs. Conditional field F1 therefore measured field-value agreement after the corresponding record had been recovered and aligned. Missing reference records and extra model records did not enter this calculation.

Some of the same field values were used both to determine record matches and to evaluate agreement within the resulting matched pairs. Conditional agreement for these fields may therefore be somewhat optimistic because greater similarity increased the likelihood that two records were paired. This is particularly relevant to `original_trait_name`, which received the largest matching weight.

For end-to-end field metrics, the same field-comparison rules were extended to unmatched records. Non-empty reference values in missing reference records contributed FN_j , while non-empty model values in extra model records contributed FP_j . End-to-end field scores therefore combined record recovery with field-value agreement. They should be interpreted alongside row-set metrics because entirely missed records lower both row-set recall and end-to-end field recall.

For each metric scope, macro field F1 was calculated as the unweighted mean of the per-field F1 scores:

$$\text{Macro Field F1} = \frac{1}{n} \sum_{j=1}^n F1_j \quad (8)$$

where $n = 21$. Macro field F1 therefore gave each of the 21 scored fields equal influence regardless of reference support (Manning et al., 2008). Reporting-frequency-weighted alternatives were not used as headline measures because common fields would dominate the summary and obscure performance on less frequently populated fields.

2.7.4 Qualitative discrepancy analysis

Automated scores were supplemented with qualitative discrepancy analysis to identify why model and reference records disagreed. Reviewed discrepancies were grouped into broad categories including missing records, unsupported extra records, supported extra records indicating possible reference omissions, field-value mismatches, unit or formatting differences, alignment problems and figure-digitisation uncertainty.

The review was used to identify recurring extraction and workflow limitations and, for the development papers, to inform later prompt revisions. It was interpretive only and was not used to change automated match labels or recalculate precision, recall or F1. All reported quantitative results therefore remained based on the automated evaluation.

2.7.5 Final prompt selection and held-out evaluation

The final prompt version was selected from the development results using conditional macro field F1 as the selection criterion. Because the intended application was human-in-the-loop curation, this criterion prioritised field agreement within recovered candidate records, while record completeness was assessed separately using row recall and end-to-end macro field F1. The selected prompt and evaluation pipeline were then locked before application to the held-out benchmark, and held-out results were not used for further prompt revision.

3 Results

Development-set results showed consistently high row precision but incomplete record recovery, with substantial variation across prompt versions, fields and papers. Results are organised in the order of the research questions. Overall extraction performance addresses research question 1, followed by the effects of prompt refinement on coverage and field quality (research question 2), the main document- and field-level sources of extraction failure (research question 3), and performance of the locked pipeline on held-out papers (research question 4). The implications of these results for human review and curation workflow (research question 5) are considered in the Discussion.

3.1 Overall extraction performance on the development benchmark

Across prompt versions, the model consistently produced a high proportion of records that could be aligned to the reference benchmark, but recovered only a minority of the complete reference record set. Across the 11 prompt versions, row precision ranged from 0.947 (v11) to 1.000 (v1), while row recall ranged from 0.102 (v1) to 0.380 (v2) (Figure 3). Conditional macro field F1, calculated only within matched record pairs, ranged from 0.551 (v1) to 0.810 (v10). End-to-end macro field F1, which additionally represented fields lost through missing records and introduced through extra records, ranged from 0.102 (v1) to 0.448 (v8). Row precision therefore remained consistently high across prompt configurations, while record recovery and field-level performance varied more substantially.

The development results identified two different operating points. v8 achieved row precision of 0.986, row recall of 0.370 and the highest end-to-end macro field F1 of 0.448, making it the coverage-oriented comparator. v10 achieved row precision of 0.975, row recall of 0.230 and the highest conditional macro field F1 of 0.810. Because conditional macro field F1 was the prespecified development-set selection criterion, v10 was selected as the final pipeline before held-out evaluation.

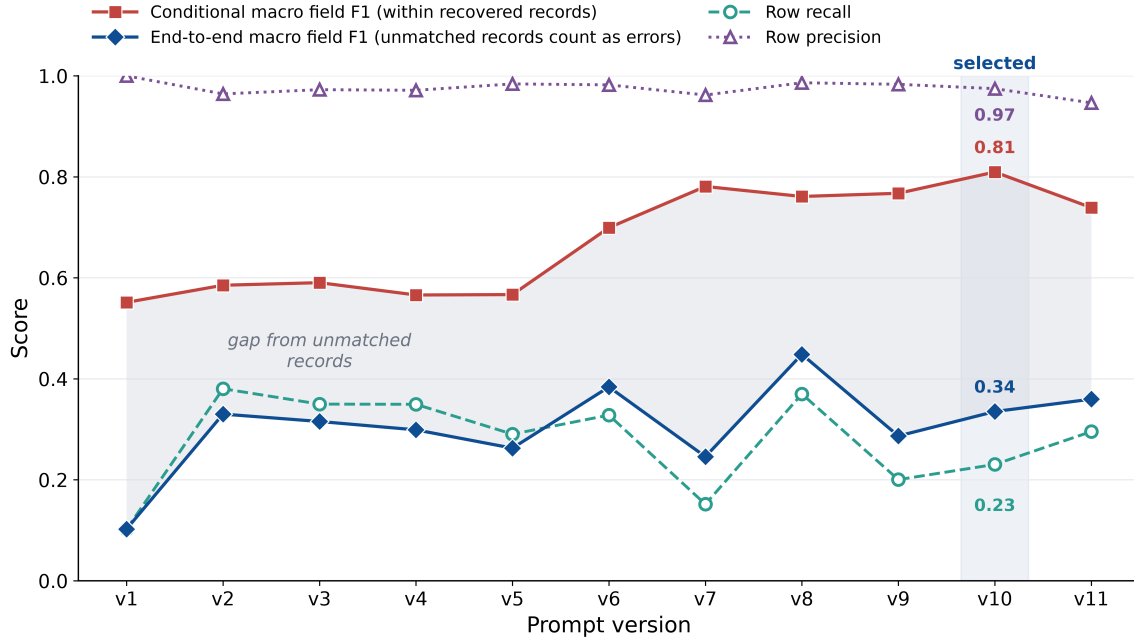


Figure 3: Extraction performance across prompt versions v1–v11 on the development benchmark. The shaded region shows the difference between conditional and end-to-end macro field F1.

3.2 Effects of prompt refinement on coverage and field quality

Prompt refinement shifted the balance between record recovery and field agreement rather than producing a monotonic improvement across all evaluation dimensions. Table 2 summarises the principal change and development performance of each prompt version.

Table 2: Prompt-version history and development-set performance across v1–v11. Records (Δ) gives the model record count and its change from the preceding version.

Ver.	Principal change and motivation	Records (Δ)	Row precision	Row recall	Cond. macro F1	End-to-end macro F1
v1	Established a minimal structured-extraction baseline	238	1.000	0.102	0.551	0.102
v2	Added field, column-role and atomic-record rules to improve cell-level completeness	921 (+683)	0.964	0.380	0.585	0.330

Continued on next page

Ver.	Principal change and motivation	Records (Δ)	Row precision	Row recall	Cond. macro F1	End-to-end macro F1
v3	Restricted extraction to focal outcomes to reduce incidental records	840 (−81)	0.973	0.350	0.590	0.316
v4	Exempted main-results tables from v3 to recover relevant excluded rows	840 (+0)	0.971	0.350	0.566	0.299
v5	Reworked figure-evidence rules to improve recovery from visual evidence	689 (−151)	0.984	0.290	0.567	0.263
v6	Restored full rules and defined error/interactor2 fields to improve field coverage	780 (+91)	0.982	0.328	0.699	0.384
v7	Removed extraction rules to test their aggregate contribution	368 (−412)	0.962	0.152	0.781	0.246
v8	Restored only one-record-per-cell guidance to test table completeness	876 (+508)	0.986	0.370	0.761	0.448
v9	Added field-source and formatting rules to reduce source ambiguity	476 (−400)	0.983	0.200	0.767	0.287
v10	Targeted grouped tables, PDF artefacts and inline figure text after discrepancy review	552 (+76)	0.975	0.230	0.810	0.335
v11	Added explicit inclusion/exclusion rules to reduce scope inconsistency	729 (+177)	0.947	0.296	0.739	0.360

The largest single-step increase in row recall occurred from v1 to v2, when explicit field definitions and atomic-record guidance were added. Across v3–v5, prompt configurations containing more restrictive or paper-specific rules were associated with lower recovery than v2. v7, which removed the extraction rules to assess their aggregate contribution, had row recall of 0.152. Restoring the table-completeness rule in v8 was associated with an increase in row recall to 0.370 and the highest end-to-end macro field F1. Subsequent versions shifted the balance between coverage and conditional field agreement, culminating in v10 achieving the highest conditional macro field F1 but lower row recall. Prompt refinement therefore changed the operating balance of the pipeline rather than producing uniform improvement across all

metrics.

Prompt effects also differed among fields rather than improving all outputs together (Figure 4). Secondary-interactor information was not successfully recovered in v2–v4 but was recovered from v6 onward, while error-unit extraction also changed markedly after v6. By v10, stage, sex, provenance and several value fields showed comparatively strong conditional F1, whereas `original_trait_name`, `original_trait_unit` and the secondary-interactor fields remained comparatively difficult. The effect of prompt refinement was therefore field-specific as well as record-level.

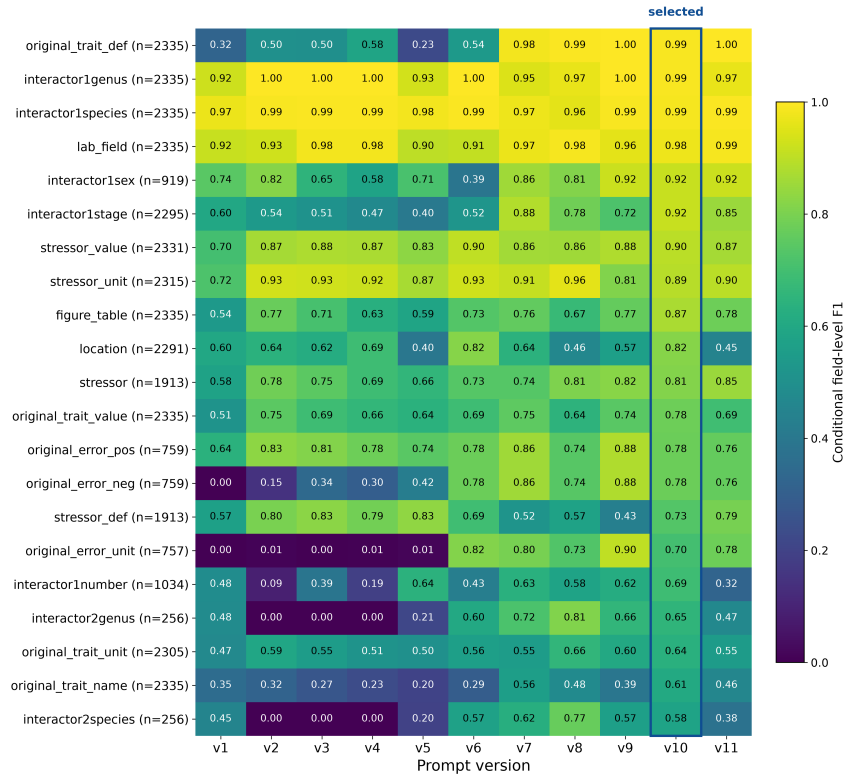


Figure 4: Conditional field-level F1 across prompt versions v1–v11 for the 21 scored fields. Fields are ordered by v10 F1, and values in parentheses indicate the number of non-empty development-reference values for each field.

3.3 Sources of extraction failure

Failure analysis considered two distinct levels: omission of complete records from a paper and disagreement in individual fields within records that were successfully recovered. Document-level omissions were characterised using v8 because it combined near-maximal row recall

with the highest end-to-end macro field F1, providing a coverage-oriented view of where records remained missing. Field-level disagreement was characterised using v10 because it was the final pipeline selected by the prespecified development-set criterion and subsequently locked for held-out evaluation.

3.3.1 Document-level record omissions

Under the coverage-oriented v8 configuration, missing records were highly concentrated in a small subset of development papers. Of 2,335 reference records, 864 were matched to v8 model output, giving row recall of 0.370 and row precision of 0.986, with 12 unmatched model records (Figure 5). Per-paper row recall had a median of 0.595 and an interquartile range of 0.168–0.912. Descriptively, median recall was 0.173 across the three figure-heavy papers, 0.595 across the three mixed-format papers and 0.688 across the nine table-heavy papers. Given the small, diversity-oriented groups, these values are descriptive rather than population-level comparisons.

Four papers were fully recovered (`paper_028`, `paper_082`, `paper_239`, `paper_298`), whereas five papers had row recall below 0.20: `paper_058` (0.081), `paper_142` (0.045), `paper_036` (0.119), `paper_096` (0.164) and `paper_262` (0.173). These five papers accounted for 1,122 of the 1,471 total missing reference records (76%), showing that omissions were concentrated rather than evenly distributed across the corpus.

The colour coding in Figure 5 separated two dominant structural failure patterns among four of the five low-recall papers. The red bars correspond to `paper_058` and `paper_142`, where difficult plot digitisation was the dominant barrier. For `paper_058`, 319 missing reference records came from Figures 2a, 2b and 4, while 342 missing records in `paper_142` came primarily from Figures 3, 4 and 8. In both cases, the relevant figures were supplied as visual evidence but many plotted values were not recovered as complete records, indicating a limitation in plot interpretation or digitisation rather than figure detection.

The blue bars correspond to `paper_036` and `paper_096`, where complex row–column table structure was the dominant barrier. `paper_036` retained values from Tables 1–3 but linearised row-header traits and column-header time points, while `paper_096` retained visually

legible tables but failed to fully expand nested cultivar, temperature and trait combinations. These omissions therefore arose from difficulties in interpreting table structure or generating the full set of records represented by the tables. The grey paper_262 showed a mixed pattern, with omissions from both Figure 2 and values reported in the Results text. Similar low-recall outcomes therefore arose from different document-structure and record-generation failures.

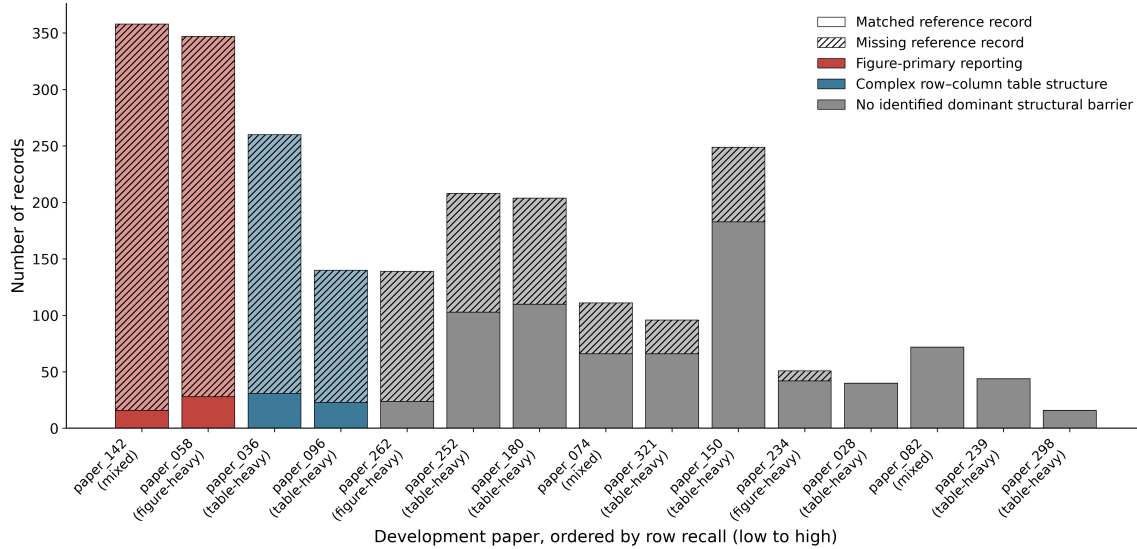


Figure 5: Development reference benchmark composition and per-paper row-level performance of v8. Each bar shows the total number of reference records, divided into records aligned to model output (solid) and unmatched reference records (hatched), with papers ordered by row recall from low to high. Reporting format is shown below each paper label. Red indicates figure-primary reporting with difficult plot digitisation, blue indicates complex row-column table structure, and grey indicates no identified dominant structural barrier. The 12 unmatched model records are not shown.

3.3.2 Field-level disagreement within recovered records

Field-level agreement also varied after records had been recovered and matched. Across the 538 matched v10 record pairs, conditional field F1 ranged from 0.582 to 0.991 across the 21 quantitatively scored fields (Figure 6). `original_trait_def`, constructed from a fixed slot sequence specified in the prompt, achieved the highest conditional F1 at 0.991. Several identity and context fields also performed strongly, including `interactor1genus` (0.989), `interactor1species` (0.985) and `lab_field` (0.976). Conversely, `stressor_def` achieved 0.734, `original_trait_unit` 0.636 and `original_trait_name` 0.606. Although both had matched-reference support of only 80 values, the two secondary-interactor fields achieved con-

ditional F1 of 0.655 for `interactor2genus` and 0.582 for `interactor2species`.

For descriptive comparison, selected fields were highlighted according to the type of extraction task they represented. Blue highlights identify examples in which the target was generated from a fixed template or had a relatively narrow and canonical output space, including `original_trait_def`, `interactor1genus`, `lab_field` and `stressor_unit`. Red highlights identify the two fields requiring more open-ended textual selection or construction, `original_trait_name` and `stressor_def`. Grey highlights show the remaining source-copied fields, for which the main task was to recover values, units, names or other information reported in the source paper. In panel (a), all four blue-highlighted fields achieved conditional F1 above 0.88, whereas both red-highlighted fields were lower. The grey fields spanned a much wider performance range. Panel (b) showed the same descriptive pattern, with the blue triangles concentrated toward high conditional precision and recall, the red squares occupying lower positions on both axes, and the grey circles distributed more broadly. Field-level difficulty was therefore heterogeneous even among records that had already been recovered.

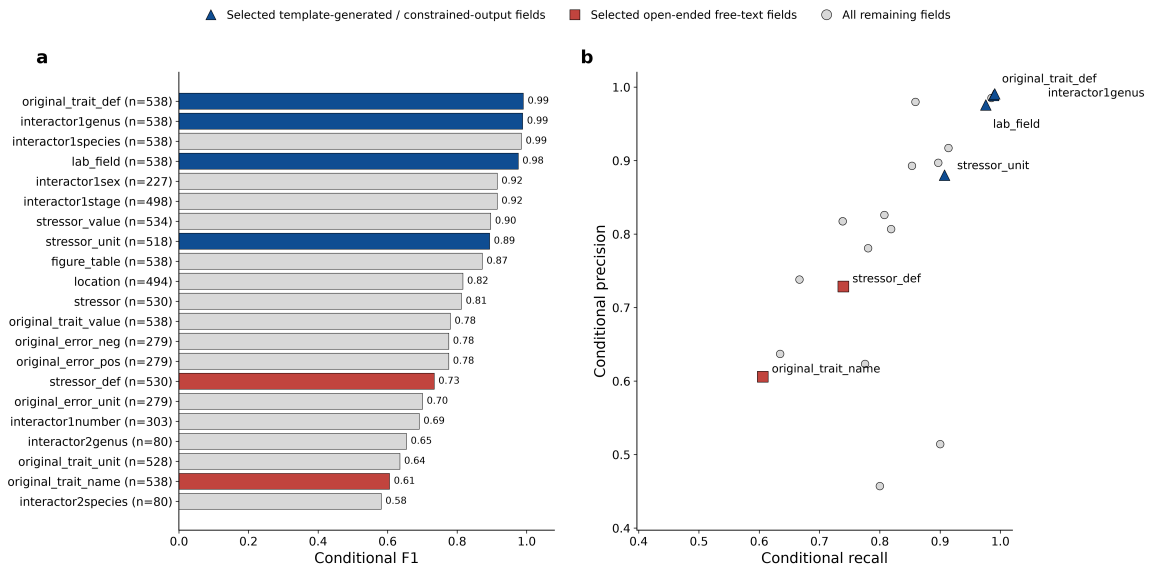


Figure 6: Conditional field-level performance of v10 across the 21 scored fields. (a) Conditional F1 by field, ordered from highest to lowest, with values in parentheses indicating matched-reference support, defined as the number of non-empty reference values among matched record pairs. (b) Conditional precision versus recall for the same fields. Blue bars and triangles highlight selected template-generated or constrained-vocabulary fields, red bars and squares highlight selected open-ended free-text fields, and grey bars and circles indicate the remaining source-copied fields.

3.4 Held-out performance of the locked v10 pipeline

The locked v10 pipeline retained a high-precision but incomplete-recovery profile on the held-out benchmark. The benchmark comprised 431 records across five papers, and v10 generated 155 records, all of which could be aligned to reference records, giving row precision of 1.000 and row recall of 0.360. Compared descriptively with development performance, row precision increased from 0.975 to 1.000 and row recall from 0.230 to 0.360. Conditional macro field F1 decreased from 0.810 to 0.729, whereas end-to-end macro field F1 increased from 0.335 to 0.394 (Table 3). Because the development and held-out benchmarks contained different papers, these differences describe performance across two paper sets rather than paired changes in the same papers. The higher held-out end-to-end score coincided with greater record recovery, while conditional field agreement was lower than on the development papers.

Metric	Development	Held-out	Change
Row precision	0.975	1.000	+0.025
Row recall	0.230	0.360	+0.129
Conditional macro field F1	0.810	0.729	-0.081
End-to-end macro field F1	0.335	0.394	+0.058

Table 3: Development and held-out performance of the locked v10 pipeline. Change is held-out minus development, calculated from unrounded values. Differences are descriptive because the two benchmarks contain different papers.

Record recovery nevertheless remained strongly paper-dependent on the held-out set. v10 recovered 80/160 records from `paper_089`, 15/28 from `paper_152`, 26/89 from `paper_231`, 5/104 from `paper_280` and 29/50 from `paper_329`. Paper-level row recall ranged from 0.048 to 0.580. Held-out evaluation therefore reproduced the main development-set pattern of very high row precision alongside incomplete and strongly document-dependent record recovery, while conditional field agreement was lower.

4 Discussion and Conclusion

This study evaluated whether a provenance-aware multimodal LLM pipeline could recover paper-reported vector-trait records at a level suitable for supporting VecTraits curation. The discussion considers overall performance, prompt refinement, document- and field-level failures, held-out transfer and the resulting role for human review before addressing limitations.

First, the pipeline was better suited to high-precision candidate generation than to complete autonomous extraction. Row precision remained consistently high, whereas record recovery was lower and more variable. In a human-in-the-loop workflow, recovered records can be checked against source evidence, but omitted records require a separate completeness check. High precision is therefore useful for limiting unmatched candidates, but insufficient for autonomous ingestion when recall remains incomplete.

Second, prompt refinement changed the balance between record coverage and field agreement rather than producing uniform improvement. v8 favoured coverage and achieved the highest end-to-end macro field F1, whereas v10 achieved the highest conditional macro field F1. Similar precision–recall trade-offs have been reported in scientific information extraction, while narrower fine-tuned tasks can achieve more balanced performance (Polak and Morgan, 2024; Dagdelen et al., 2024). These studies are not directly comparable to full-document multimodal extraction, but they support the broader observation that performance depends on task scope and output constraints. The lower recall observed here may partly reflect the greater difficulty of reconstructing multi-field records from heterogeneous text, tables and figures.

Third, incomplete recovery was strongly structure-dependent. In figure-primary papers, relevant figures were available but plotted values were often not reconstructed as complete records, implicating visual interpretation and digitisation. In complex tables, nested row–column combinations were not consistently expanded into complete records, implicating structural interpretation and record assembly. These mechanisms require different interventions: specialised plot digitisation for figures and structure-aware preprocessing or explicit header relationships for complex tables. This interpretation is consistent with studies showing benefits from explicit table structure and specialised curve-extraction components (Bai et al., 2024; Khalighinejad et al., 2025). Generic prompt refinement alone may therefore be insufficient

for difficult document structures, and broad labels such as table-heavy or figure-heavy do not identify the specific structural barrier. If such structures cluster by trait family or study design, omissions could also create structured missingness in downstream trait datasets, although this was not tested here.

At the field level, difficulty was not explained by a simple structured-versus-free-text distinction. `original_trait_def`, generated from a fixed slot sequence, achieved very high agreement, whereas `original_trait_name` was harder because it required selecting an appropriate author-used term from several possible expressions and applying predefined naming rules. `stressor_def` similarly required semantic selection rather than simple value recovery. Fields with clearer expected forms generally performed better than those requiring greater selection or normalisation judgement. Direct recovery from the paper therefore did not itself guarantee high agreement, suggesting that semantically selective fields warrant more explicit rules or targeted curator review.

Fourth, held-out evaluation showed high row precision on held-out papers, but field agreement and record recovery did not generalise uniformly. All v10 outputs aligned to held-out reference records, while recovery remained incomplete and paper-dependent. Conditional macro field F1 decreased, whereas end-to-end macro field F1 increased because more reference records were recovered. The higher end-to-end score therefore reflected greater coverage rather than stronger within-record agreement, reinforcing that different components of performance can transfer differently across papers.

Finally, the observed profile supports a human-in-the-loop workflow in which the model generates provenance-linked candidate records while curators retain responsibility for verification and completeness. Candidate records would undergo automated validation and source verification, with closer review of semantically ambiguous fields and a separate completeness pass before database entry. Figure-primary records may require plot digitisation, while complex tables may require structure-aware preprocessing or manual extraction. This preserves the distinction between source-faithful extraction and later semantic harmonisation, consistent with MIREVTD and FAIR principles (Ryan et al., 2026; Wilkinson et al., 2016).

Two study-design limitations affect generalisation. Prompt versions sometimes changed

multiple instructions simultaneously, so differences cannot be attributed causally to individual rules. This reflected iterative development aimed at correcting recurrent error combinations, but controlled future experiments should vary prompt components incrementally. Reusing the same 15 development papers was also necessary for iterative refinement but may have caused later prompts to become partly tailored to characteristics of the development corpus. Locking v10 before held-out evaluation reduced this risk, while a larger untouched benchmark would provide a stronger test. The five-paper held-out set was also small because complete record-level benchmarks are labour-intensive and the study prioritised structural diversity. Its results therefore demonstrate transfer to held-out papers rather than precise wider-literature performance, which requires a larger independent corpus.

Additional limitations concern the model and evaluation framework. Only one hosted multimodal model was evaluated. Holding the model constant focused the study on prompt and pipeline behaviour, but some findings may be model-specific. Applying the locked prompt unchanged to other models would directly test cross-model transferability. Repeated generations were not evaluated, so repeated locked-pipeline runs are needed to quantify stochastic variability. Record-level scores also depend on matching because model and reference outputs lack shared identifiers. Matching choices, and reuse of identifying fields in conditional field evaluation, may therefore influence scores. Sensitivity analyses using alternative matching specifications, including leave-one-field-out alignment for key variables, would test robustness. Finally, curator time and correction burden were not measured because the study evaluated extraction quality rather than operational efficiency. A future manual-only comparison should measure these practical outcomes.

Overall, multimodal LLM extraction can support first-pass vector-trait curation, but the present results do not justify autonomous ingestion. The locked v10 pipeline retained very high held-out row precision while recovering only 36.0% of reference records, leaving incomplete and structure-dependent recovery as a principal bottleneck. The most defensible use is provenance-linked candidate generation with expert verification and a separate completeness check. Future progress should prioritise structure-specific document processing, cross-model transfer, larger independent evaluation and practical testing with curators.

Data and Code Availability

The code, versioned prompts, extraction schemas, corpus manifests, matching and evaluation scripts, archived model outputs, and manually curated development and held-out reference benchmarks supporting this study are available in the versioned GitHub release at <https://github.com/yukiwxq-ui/VecTraits-LLM-Extraction/releases/tag/v1.0.1>. A citable archival copy of this release is preserved on Zenodo at <https://doi.org/10.5281/zenodo.22005129>. The archived package is licensed under CC BY 4.0. Source-paper PDFs are not redistributed because they remain subject to publisher access and reuse terms.

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Appendix A: Corpus composition

Table A1: Selected 20-paper corpus used for prompt development and evaluation. The table summarises the development or held-out split, existing VecTraits row count, inferred primary trait family, inferred reporting format, dataset source and selection rationale. Trait family and reporting format were used as sampling descriptors rather than definitive biological classifications.

Paper ID	Split	VecTraits rows	Primary trait family	Reporting format	Dataset source	Selection rationale
paper_028	development	12	behaviour_movement	table-heavy	537	Selected as the highest-row representative of the behaviour/movement primary trait family.
paper_036	development	229	development	table-heavy	653	Selected by the diversity score to add a new dataset source and citation group while retaining a high row count.
paper_058	development	326	survival_mortality	figure-heavy	505	Selected as the highest-row representative of the survival/mortality primary trait family.
paper_074	development	2840	development	mixed	1228	Selected as the highest-row remaining representative of mixed-format reporting.
paper_082	development	272	development	mixed	1513	Selected by the diversity score to add a new dataset source and citation group while retaining a high row count.
paper_096	development	280	development	table-heavy	318	Selected as the highest-row remaining representative of table-heavy reporting.
paper_142	development	263	longevity	mixed	564	Selected by the diversity score to add a new dataset source and citation group while retaining a high row count.
paper_150	development	238	development	table-heavy	354	Selected by the diversity score to add a new dataset source and citation group while retaining a high row count.
paper_180	development	234	development	table-heavy	113	Selected by the diversity score to add a new dataset source and citation group while retaining a high row count.

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Table A1: Selected 20-paper corpus used for prompt development and evaluation. Continued.

Paper ID	Split	VecTraits rows	Primary trait family	Reporting format	Dataset source	Selection rationale
paper_234	development	41	physiology_environment	figure-heavy	539	Selected as the highest-row representative of the physiology/environmental response primary trait family.
paper_239	development	40	body_size_morphology	table-heavy	391	Selected as the highest-row representative of the body size/morphology primary trait family.
paper_252	development	208	development	table-heavy	481	Selected by the diversity score to add a new dataset source and citation group while retaining a high row count.
paper_262	development	343	longevity	figure-heavy	634	Selected as the highest-row remaining representative of figure-heavy reporting.
paper_298	development	16	feeding_behaviour	table-heavy	661	Selected as the highest-row representative of the feeding behaviour primary trait family.
paper_321	development	216	development	table-heavy	1529	Selected by the diversity score to add a new dataset source and citation group while retaining a high row count.
paper_089	held-out	7908	development	mixed	223	Selected as the highest-row representative of the development primary trait family and assigned to the held-out split to retain evaluation diversity.
paper_152	held-out	356	other_uncategorised	table-heavy	436	Selected as the highest-row representative of the other/uncategorised primary trait family and assigned to the held-out split to retain trait-family diversity.
paper_280	held-out	63	longevity; reproduction_fecundity	table-heavy	1019	Pre-specified replacement for the excluded supplementary-dependent longevity paper; selected as the highest-ranked accessible unused longevity candidate before held-out model evaluation.
paper_231	held-out	7	development	text-heavy	856	Selected as the highest-row remaining representative of text-heavy reporting and assigned to the held-out split to retain reporting-format diversity.

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Table A1: Selected 20-paper corpus used for prompt development and evaluation. Continued.

Paper ID	Split	VecTraits rows	Primary trait family	Reporting format	Dataset source	Selection rationale
paper_329	held-out	1024	reproduction_fecundity	figure-heavy	965	Selected as the highest-row representative of the reproduction/fecundity primary trait family and assigned to the held-out split to retain evaluation diversity.

Appendix B: Extraction schema definitions

Table B1: VecTraits extraction schema field definitions.

Field	Plain English name	Definition	Example
Model extraction target fields			
location	Study location	Geographic location where organisms were collected or the experiment was conducted; colony or rearing-facility descriptions are excluded.	Kampala, Uganda
lab_field	Study setting	One of laboratory, field or semi-field.	laboratory
original_trait_name	Trait name	The shortest name the authors themselves used for this variable anywhere in the full text (title, abstract, methods, results or discussion); modifiers such as “mean” or “relative” are stripped when the bare form also appears in the paper, and author abbreviations are expanded to the full form found in the text. This is a selection-and-normalisation rule, not a verbatim copy of a single occurrence such as a table column header.	development time
original_trait_def	Trait definition	A single sentence synthesised from a fixed slot order: species and life stage, trait name (with a parenthetical for technical or abbreviated terms), sample basis (mean/median of <i>N</i> individuals or an individual measurement), and the stressor condition. Not quoted verbatim from the source paper.	<i>Brevicoryne brassicae</i> nymphal developmental time from egg hatch to adult eclosion, mean of 30 individuals, at 15°C
original_trait_value	Trait value	The reported numeric or categorical value only.	13.8
original_trait_unit	Trait unit	The unit as written in the paper; inverse units are written with a trailing -1 (e.g. mg day ⁻¹) rather than ⁻¹ .	days
original_error_pos	Positive error	Positive uncertainty component from “mean ± X” or an upper asymmetric bound.	0.4
original_error_neg	Negative error	Negative uncertainty component (equal to original_error_pos when the reported error is symmetric).	0.4
original_error_unit	Error type	The type of uncertainty reported, restricted to SE, SD, SEM, 95% CI, range or quartiles. This is not the trait measurement unit.	SE

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Table B1: VecTraits extraction schema field definitions. continued

Field	Plain English name	Definition	Example
interactor1genus	Primary genus	Genus of the organism whose trait is being measured.	Aedes
interactor1species	Primary species	Species epithet only; strain, lineage or phenotype descriptors are omitted.	aegypti
interactor1stage	Primary life stage	Life stage as reported (e.g. egg, larva, nymph, pupa, adult, first instar); juvenile is used when the measurement spans multiple pre-adult stages. Life stage only, not sex or a developmental process.	larva
interactor1sex	Primary sex	One of female, male or mixed; null when not reported.	female
interactor1number	Number of primary organisms	Number of individuals or replicates per treatment condition, where reported.	30
interactor2genus	Secondary genus	Genus of a second interacting organism (host, prey, pathogen, parasitoid or host plant), where relevant; null for single-organism studies.	Plasmodium
interactor2species	Secondary species	Species epithet of interactor2, where relevant.	falciparum
stressor	Stressor or treatment name	The main experimental or environmental treatment (e.g. temperature, humidity, diet, insecticide, host type). When multiple stressors were reported for the same record, temperature is recorded here and the remaining conditions are recorded in notes.	temperature
stressor_def	Stressor definition	The paper's own description of the stressor condition.	constant rearing temperature
stressor_value	Stressor value	The reported condition value.	28
stressor_unit	Stressor unit	The reported condition unit.	°C
figure_table	Source figure or table	The table or figure number that is the primary source for the record (e.g. "Table 2", "Figure 1A"); climate/environmental background tables and statistical summary figures are excluded as sources.	Table 2
notes	Extraction notes	Additional context not captured elsewhere: secondary stressors (formatted as "Other stressor(s): X"), asymmetric error bounds, or caveats. Values already captured in other fields are not repeated here.	Other stressor(s): relative humidity 70%
Linking field, not a model extraction target			

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Table B1: VecTraits extraction schema field definitions. continued

Field	Plain English name	Definition	Example
paper_id	Paper identifier	Stable internal identifier assigned to each source paper in this study. This field was added during corpus assembly and was not extracted by the model.	paper_010

Table B2: Field-specific normalisation and comparison rules used during evaluation.

Field(s)	Normalisation/comparison rule
Missing-value tokens	Empty string, na, n/a, null, none, nan, not reported, not stated, nr and - (case-insensitive) are mapped to empty before comparison.
Default text fields	Lower-cased, whitespace collapsed; punctuation retained unless a field-specific rule removes it.
original_trait_name; interactor1stage	Lower-cased, whitespace collapsed, punctuation removed.
original_trait_unit; original_error_unit; stressor_unit	Lower-cased, then mapped through a unit alias table (e.g. °C, degrees Celsius and C → celsius; d, day → days).
interactor1stage	Additionally mapped through a life-stage alias table (e.g. larvae, larval → larva).
interactor1sex	Mapped through a sex alias table (e.g. m → male; both sexes → mixed).
lab_field	Mapped through an alias table onto exactly one of laboratory, field or semi-field.
original_trait_value; original_error_pos; original_error_neg; stressor_value	Parsed to a numeric value and compared within 1% relative tolerance; falls back to string comparison if either value is non-numeric.
interactor1number	Parsed to a numeric value and compared for an exact match (0% tolerance), since organism counts are not expected to vary.
location; original_trait_def; stressor_def	Relaxed matching applied when the exact normalised strings differ: counted as a match if one value contains the other, or if the two share at least two tokens of four or more characters.